Non-Uniform Accn Worksheet

Class 11 Physics | Nine Education IIT Academy | Kinematics — Variable Acceleration

Name: ______________________ Date: ______________

The Three Working Relations

1	**$v = dx/dt$**	Links position and velocity directly — use to move between $x(t)$ and $v(t)$.
2	**$a = dv/dt$**	Use when acceleration is a function of **time**: $a = f(t)$. Integrate over t.
3	**$a = v\cdot(dv/dx)$**	Use when acceleration is a function of **position**: $a = f(x)$. No t needed — separate v dv from a(x) dx and integrate with limits.

Take all quantities in SI units. Where a numeric answer is irrational, leave it in surd form or round to 2 decimal places.

Formula 1: $v = dx/dt$ — Position ↔ Velocity

1. The position of a particle is $x = 4t^2 - 3t + 2$ (m). Find its velocity and acceleration as functions of time.

2. A particle moves with $v(t) = 6t^2 - 4t$ (m/s). If $x = 3$ m at $t = 0$, find its position at $t = 2$ s.

3. The position of a particle is $x = t^3 - 6t^2 + 9t$ (m). Find the time(s) at which the particle is momentarily at rest.

4. The position of a particle is $x = 5t^2 - 2t^3$ (m). Find the velocity at $t = 1$ s, and the time ($t > 0$) at which the velocity becomes zero.

5. A particle's velocity is $v = 4t + 1$ (m/s). If $x = 2$ m at $t = 1$ s, find its position at $t = 3$ s.

6. The displacement of a particle is $x = 2t^3 - 5t + 1$ (m). Find its acceleration at $t = 2$ s.

7. The position of a particle is $x = t^2 - 4t + 4$ (m). Find the time at which the particle is at the origin, and its velocity at that instant.

8. The position of a particle is $x = 3t^2 + 2t$ (m). Find its velocity at $t = 4$ s.

9. A particle's velocity is $v(t) = 2t + 5$ (m/s), and it starts at the origin ($x = 0$ at $t = 0$). Find its position at $t = 4$ s.

10. The position of a particle is $x = t^3 - 3t^2$ (m). Find the acceleration at $t = 2$ s.

Formula 2: $a = dv/dt$ — When $a = f(t)$

11. A particle starts from rest at the origin with acceleration $a = 4t$ (SI). Find its velocity and position at $t = 3$ s.

12. A particle has acceleration $a = (6 - 2t)$ m/s² and starts with $u = 0$ at $t = 0$ from the origin. Find the time at which its velocity is maximum, and that maximum velocity.

13. A particle starts with $u = 5$ m/s and acceleration $a = 3t^2$ (SI). Find its velocity at $t = 2$ s and its displacement in that time (starting at $x = 0$).

14. A particle experiences acceleration $a = 2 + 6t$ (SI), starting from rest at $x = 2$ m. Find the velocity and position at $t = 2$ s.

15. The acceleration of a particle is $a = (6t - 6)$ m/s², and $v = 0$ at $t = 0$. Find $v(t)$, the other time ($t > 0$) at which the particle is momentarily at rest, and its velocity at $t = 1$ s.

16. A particle moves with acceleration $a = (4 - t^2)$ m/s², starting from rest at the origin. Find its velocity and position at $t = 2$ s.

17. A particle's acceleration is $a = (2t + 3)$ m/s². It starts with $u = 2$ m/s at $t = 0$ from $x = 0$. Find the velocity at $t = 3$ s and the distance covered in that time.

18. The acceleration of a particle varies as $a = (3t^2 - 4t)$ m/s². If it starts from rest at the origin, find its displacement between $t = 1$ s and $t = 3$ s.

19. A rocket accelerates as $a = (20 - 4t)$ m/s² from rest at ground level. Find the time at which its velocity is maximum, the maximum velocity, and the height reached at that time.

20. A particle starts from rest at the origin with acceleration $a = 6t$ (SI). Find its velocity and position at $t = 2$ s.

21. A particle has initial velocity $u = 3$ m/s and acceleration $a = 2t$ (SI). Find its velocity at $t = 3$ s.

22. The acceleration of a particle is $a = (10 - 2t)$ m/s², starting from rest. Find the time at which the velocity is maximum, and that maximum velocity.

23. A particle moves with acceleration $a = 3t^2$ (SI), starting from rest at the origin. Find its position at $t = 2$ s.

Formula 3: $a = v(dv/dx)$ — When $a = f(x)$

24. A particle has acceleration $a = 9x$ (SI). Starting from rest at $x = 0$, find its speed at $x = 2$ m.

25. A particle has acceleration $a = -8x$ (SI), with $v = 0$ at $x = 3$ m. Find its speed at $x = 1$ m.

26. A particle experiences acceleration $a = 6x + 4$ (SI) and has speed 3 m/s at $x = 0$. Find its speed at $x = 2$ m.

27. A particle has acceleration $a = -2x$ (SI). At $x = 0$, its speed is 4 m/s. Find its speed at $x = 1$ m.

28. A particle starts from rest at the origin with acceleration $a = 6x^2$ (SI). Find its speed when it reaches $x = 1$ m.

29. A block released from rest at $x = 0$ experiences acceleration $a = 2x$ (SI). Find its speed when it reaches $x = 4$ m.

30. A particle's velocity is related to its position by $v^2 = 16 - 4x^2$ (SI). Find its acceleration as a function of x.

31. A particle is momentarily at rest at $x = 0$ and has acceleration $a = (8 - 2x)$ m/s². Find its speed at $x = 2$ m.

32. The velocity of a particle varies with position as $v = 2x$ (SI). Find the acceleration as a function of x, and its value at $x = 3$ m.

33. A particle has acceleration $a = 4x$ (SI), and its speed is 0 at $x = 0$. Find its speed at $x = 3$ m.

34. A particle has acceleration $a = -9x$ (SI). It has speed 10 m/s at $x = 0$. Find its speed at $x = 2$ m.

35. A particle starts from rest at $x = 1$ m with acceleration $a = 8x$ (SI). Find its speed at $x = 3$ m.

Hard / Challenge

36. A particle moves with acceleration $a = -kv$ (resistive), starting with velocity v_0 at $t = 0$. Show that $v = v_0e^{-kt}$, and find the time at which the velocity drops to half of v_0, in terms of k.

37. A particle's acceleration is $a = 2x + 3$ (SI). Starting from rest at $x = 0$, find its speed at $x = 2$ m.

38. The velocity of a particle varies with position as $v = 3x^2$ (SI). Find the acceleration as a function of x, and evaluate it at $x = 2$ m.

39. A particle's position is $x = t^3 - 6t^2 + 9t$ (m) — the same function as Q3. Find the total distance travelled between $t = 0$ s and $t = 4$ s (the particle reverses direction during this interval).

40. A particle's acceleration varies with position as $a = (-4x + 8)$ m/s², and it is momentarily at rest at $x = 1$ m. Show that it is also momentarily at rest at $x = 3$ m, and find its maximum speed and the position at which it occurs.

41. A particle's acceleration is $a = -kx$ (SI, $k > 0$), and it has speed v_0 at $x = 0$. Show that its speed at any position is $v^2 = v_0^2 - kx^2$, and hence find the maximum displacement from the origin, in terms of v_0 and k.

42. A particle's acceleration is $a = 2v$ (SI) — it depends on velocity. If $v = 1$ m/s at $t = 0$, find the time taken for the velocity to become 4 m/s. (Hint: use $a = dv/dt$ and separate variables.)

43. A particle's velocity varies with position as $v = k\sqrt{x}$ (k constant). Show that its acceleration is constant, and find its value in terms of k.

44. A particle starts from rest at the origin with acceleration $a = 12t^2$ (SI). Find its position as a function of time, and the time at which it has travelled 16 m and its velocity at that instant.

45. A particle's acceleration is $a = -\omega^2 x$ (SI, ω constant). It starts at $x = A$ with $v = 0$. Using $a = v(dv/dx)$, show that its speed at the origin is $v = \omega A$.

46. A particle's velocity is $v = 3t^2 - 18t + 24$ (m/s), and $x = 0$ at $t = 0$. Find the total distance travelled by the particle between $t = 0$ s and $t = 5$ s.

47. A particle's velocity and position are related by $v^2 = 4 + 6x$ (SI). Show that the particle's acceleration is constant, and find its value.

48. A particle experiences acceleration $a = 3x^2 - 2$ (SI). It starts from rest at $x = 0$. Find its speed at $x = 2$ m.

49. A particle starts from rest at the origin with acceleration $a = 4t^3$ (SI). Find v(t) and x(t), and hence find the time at which v(t) equals x(t).

50. A particle's acceleration is $a = -2x + 6$ (SI). It is momentarily at rest at $x = 1$ m and at one other position. Find that other position and the particle's maximum speed, along with the position where it occurs.

JEE Mains PYQ's

Numbers and wording below are rewritten, not copied verbatim, from the cited official paper — each question is built on the same underlying physics as the referenced JEE Main/AIEEE PYQ.

51. A particle's position obeys $x^2 = pt^2 + 2qt + r$ (p, q, r are constants, SI units). If the particle's acceleration varies with its position as $a \propto x^{-n}$, find the value of n.
(A) 1 (B) 2 (C) 3 (D) 4
Based on JEE Main 2020, 9 Jan Morning Slot

52. A particle moving with an initial speed of 9 m/s decelerates at a rate given by $dv/dt = -3\sqrt{v}$ (v in m/s, t in s). The time taken for the particle to come to rest is:
(A) 1 s (B) 2 s (C) 3 s (D) 6 s
Based on AIEEE 2011

53. The position of a particle moving along the x-axis is given by $x = 2t^3 - 9t^2 + 12t$ (m). The velocity of the particle at the instant its acceleration becomes zero is:
(A) −1.5 m/s (B) 3 m/s (C) −4.5 m/s (D) 0 m/s
Based on JEE Main 2024, 29 Jan Evening Shift

54. A particle starts from rest at the origin and moves along the x-axis such that its speed varies with position as $v = 5\sqrt{x}$ (SI units). Its acceleration is:
(A) 5 m/s² (constant) (B) 12.5 m/s² (constant) (C) 25/x m/s² (D) 25√x m/s²
Based on JEE Main 2024, 1 Feb Evening Shift

55. The velocity of a particle varies with its displacement as $v = \sqrt{1800 + 8x}$ (SI units). The acceleration of the particle is:
(A) 2 m/s² (B) 4 m/s² (C) 8 m/s² (D) 16 m/s²
Based on JEE Main 2021, 27 Aug Morning Shift

56. A particle starts from the origin (x = 0, t = 0) and moves along the positive x-axis with speed $v = 3\sqrt{x}$ (SI units). Its speed at t = 2 s is:
(A) 3 m/s (B) 4.5 m/s (C) 6 m/s (D) 9 m/s
Based on JEE Main 2019, 12 Apr Evening Slot

57. The position of a particle moving along the x-axis is $x(t) = pt + qt^2 - rt^3$, where p, q, r are positive constants. When the particle's acceleration becomes zero, its velocity is:
(A) $p + q^2/(3r)$ (B) $p + q^2/r$ (C) $p + 4q^2/(3r)$ (D) $p - q^2/(3r)$
Based on JEE Main 2019, 9 Apr Evening Slot

58. If the relation between time t and position x of a particle moving along a straight line is $t = \sqrt{x} + 3$, the velocity of the particle at t = 5 s is:
(A) 1 m/s (B) 2 m/s (C) 4 m/s (D) 8 m/s
Based on JEE Main 2022, 29 Jul Morning Shift

Non-Uniform Accn Worksheet — Answer Key

Formula 1 (Q1–10)

1. $v = (8t - 3)$ m/s; $a = 8$ m/s² (constant)

2. $x(2) = 11$ m

3. $t = 1$ s and $t = 3$ s

4. $v(1) = 4$ m/s; velocity $= 0$ at $t = 5/3$ s

5. $x(3) = 20$ m

6. $a(2) = 24$ m/s²

7. $t = 2$ s (a double root — the particle just touches the origin); $v(2) = 0$

8. $v(4) = 26$ m/s

9. $x(4) = 36$ m

10. $a(2) = 6$ m/s²

Formula 2 (Q11–23)

11. $v(3) = 18$ m/s, $x(3) = 18$ m

12. $t = 3$ s, $v_{max} = 9$ m/s

13. $v(2) = 13$ m/s, $x(2) = 14$ m

14. $v(2) = 16$ m/s, $x(2) = 14$ m

15. $v(t) = 3t^2 - 6t$; rest again at $t = 2$ s; $v(1) = -3$ m/s

16. $v(2) = 16/3 \approx 5.33$ m/s; $x(2) = 20/3 \approx 6.67$ m

17. $v(3) = 20$ m/s; $x(3) = 28.5$ m

18. $\Delta x = 8/3 \approx 2.67$ m

19. $t = 5$ s, $v_{max} = 50$ m/s, height $= 500/3 \approx 166.7$ m

20. $v(2) = 12$ m/s, $x(2) = 8$ m

21. $v(3) = 12$ m/s

22. $t = 5$ s, $v_{max} = 25$ m/s

23. $x(2) = 4$ m

Formula 3 (Q24–35)

24. $v = 6$ m/s

25. $v = 8$ m/s

26. $v = 7$ m/s

27. $v = \sqrt{14} \approx 3.74$ m/s

28. $v = 2$ m/s

29. $v = 4$ m/s

30. $a = -4x$ m/s²

31. $v = 2\sqrt{6} \approx 4.90$ m/s

32. $a = 4x$; $a(3) = 12$ m/s²

33. $v = 6$ m/s

34. $v = 8$ m/s

35. $v = 8$ m/s

Hard / Challenge (Q36–50)

36. $t = (\ln 2)/k$

37. $v = 2\sqrt{5} \approx 4.47$ m/s

38. $a = 18x^3$; $a(2) = 144$ m/s²

39. total distance = 12 m (net displacement = 4 m)

40. rest confirmed at $x = 3$ m; maximum speed = 2 m/s, occurring at $x = 2$ m

41. $v^2 = v_0^2 - kx^2$; maximum displacement $x_{max} = v_0/\sqrt{k}$

42. $t = (1/2) \ln 4 = \ln 2 \approx 0.69$ s

43. $a = v(dv/dx) = k^2/2$ (constant)

44. $x(t) = t^4$; $t = 2$ s when $x = 16$ m, with $v(2) = 32$ m/s

45. $v^2 = \omega^2 A^2 \Rightarrow v = \omega A$ at $x = 0$

46. total distance = 28 m (net displacement = 20 m)

47. $a = 3$ m/s² (constant, independent of x)

48. $v = 2\sqrt{2} \approx 2.83$ m/s

49. $v(t) = t^4$, $x(t) = t^5/5$; $v(t) = x(t)$ at $t = 5$ s

50. other rest position: $x = 5$ m; maximum speed $= 2\sqrt{2} \approx 2.83$ m/s, occurring at $x = 3$ m

JEE Mains PYQ's (Q51–58)

51. (C) $n = 3$

52. (B) $t = 2$ s

53. (A) -1.5 m/s

54. (B) 12.5 m/s²

55. (B) 4 m/s²

56. (D) 9 m/s

57. (A) $p + q^2/(3r)$

58. (C) 4 m/s